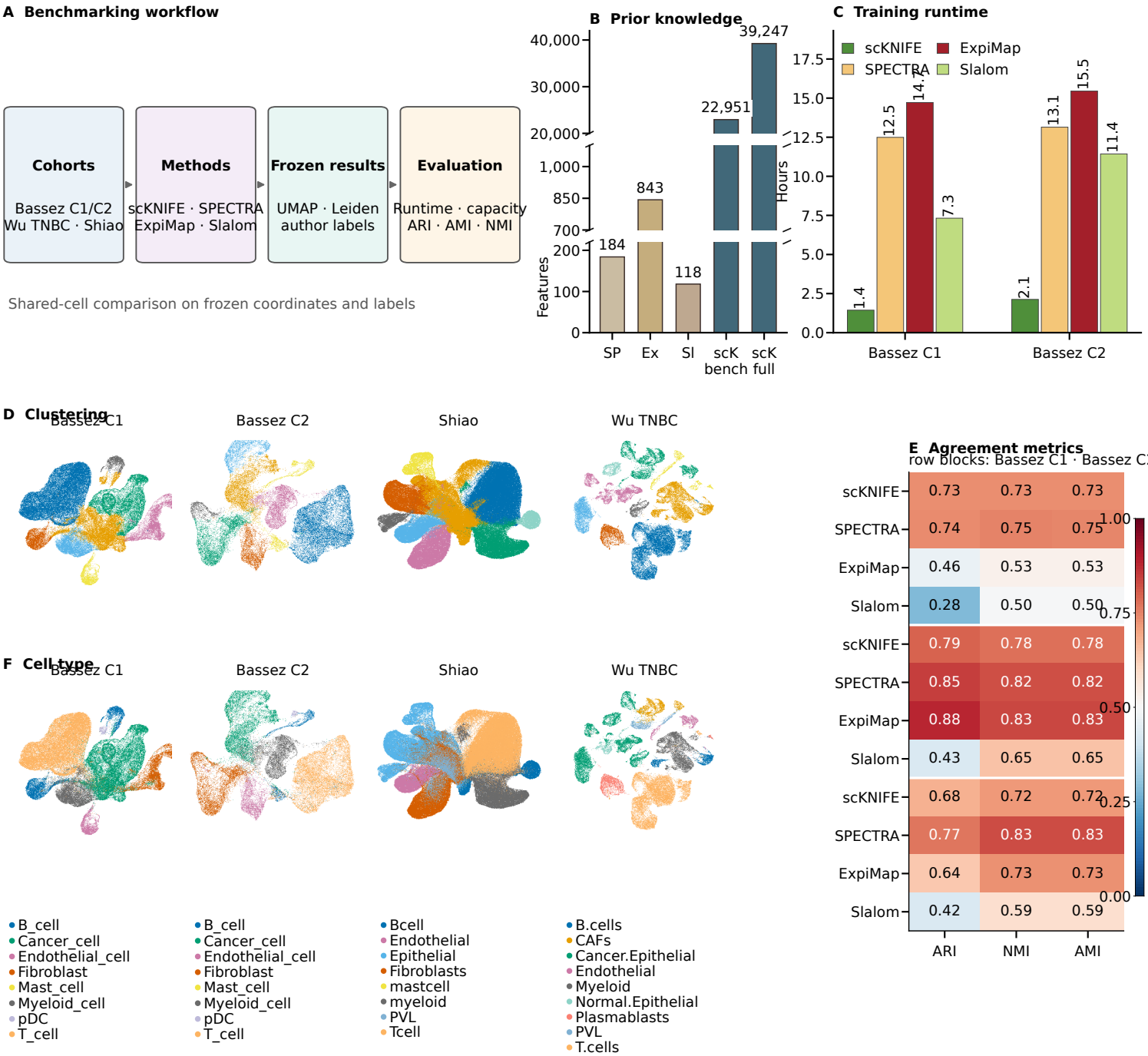

A Benchmarking workflow
Cohorts
Bassez C1/C2
Wu TNBC · Shiao
Methods
scKNIFE · SPECTRA
ExpiMap · Slalom
Frozen results
UMAP · Leiden
author labels
Evaluation
Runtime · capacity
ARI · AMI · NMI
Shared-cell comparison on frozen coordinates and labels
B Prior knowledge
Features
Hours
184
843
118
22,951
39,247
SP
Ex
Sl
scK bench
scK full
C Training runtime
scKNIFE
ExpiMap
SPECTRA
Slalom
1.4
12.5
14.5
7.3
2.1
13.1
15.5
11.4
Bassez C1
Bassez C2
D Clustering
Bassez C1
Bassez C2
Shiao
Wu TNBC
E Agreement metrics
row blocks: Bassez C1 · Bassez C2 · W
scKNIFE 0.73 0.73 0.73
SPECTRA 0.74 0.75 0.75
ExpiMap 0.46 0.53 0.53
Slalom 0.28 0.50 0.50
scKNIFE 0.79 0.78 0.78
SPECTRA 0.85 0.82 0.82
ExpiMap 0.88 0.83 0.83
Slalom 0.43 0.65 0.65
scKNIFE 0.68 0.72 0.72
SPECTRA 0.77 0.83 0.83
ExpiMap 0.64 0.73 0.73
Slalom 0.42 0.59 0.59
ARI
NMI
AMI
F Cell type
Bassez C1
Bassez C2
Shiao
Wu TNBC
B_cell
Cancer_cell
Endothelial_cell
Fibroblast
Mast_cell
Myeloid_cell
pDC
T_cell
B_cell
Cancer_cell
Endothelial_cell
Fibroblast
Mast_cell
Myeloid_cell
pDC
T_cell
Bcell
Endothelial
Epithelial
Fibroblasts
mastcell
myeloid
PVL
Tcell
B.cells
CAFs
Cancer.Epithelial
Endothelial
Myeloid
Normal.Epithelial
Plasmablasts
PVL
T.cells
G TP53 · Doxo | scKNIFE
Pearson r = 0.65
Spearman ρ = 0.54
G TP53 · Doxo | Scanpy
Pearson r = 0.63
Spearman ρ = 0.40
H PIK3R1 · Afi | scKNIFE
Pearson r = 0.76
Spearman ρ = 0.70
H PIK3R1 · Afi | Scanpy
Pearson r = 0.74
Spearman ρ = 0.70
Score
TP53
PIK3R1